THE UNIVERSITY OF BRITISH COLUMBIA
Faculty of Forestry

Dear Dr. Berenbaum,

Please consider our revised manuscript, "Solstice optimizes thermal growing season" (#2025-06796), for publication as a Brief Report in *PNAS*.

This work addresses the recent hypothesis that the summer solstice may be a universal cue for major plant physiological transitions.[1,2], which—if correct—could redefine how we predict plant responses to climate change. Using a novel thermal trade-off framework, we test the utility of solstice as a cue over past, present, and future climates across Europe and North America. Our results show that the solstice often aligns with a thermal optimum, but also finds substantial regional variation that raises questions about how adaptive a solstice cue may be across space. This work brings a new perspective to the solstice debate, with implications for both forecasting plant responses to global warming and answering fundamental biological questions.

We appreciated that both reviewers and the editor recognized the significance of our findings and made helpful comments to improve the manuscript. In addressing these comments, we have made edits throughout the text, included new analyses to show our results extend to North America, and updated both main text figures to include these new results. Additionally, we are now more careful with our terminology and in explaining the current hypotheses and we have worked to clearly acknowledge the limitations of our approach—especially regarding other climatic constraints, such as moisture availability, that influence the growing season. Altogether we believe that these changes based on the reviewers' feedback have led to a stronger contribution bridging across research fields from physiology, ecology, and climatology.

We detail our changes, point-by-point, in the following pages. Both authors have contributed to and approved of this submission, and the manuscript is not currently submitted to or in review in an other journal. We hope that you will find it suitable for publication in *PNAS*.

Sincerely,

V. Van der Meersch

Victor Van der Meersch, PhD
Forest & Conservation Sciences
University of British Columbia

References

[1] C. M. Zohner et al. *Science* (2023).

[2] V. Journé et al. *Nature Plants* (2024).